BMC Public Health

RESEARCH

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Long-term trends of influenza-like illness and severe acute respiratory infection across pandemic phases: an interrupted time-series study

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Abstract

Background Influenza-like illness (ILI) and severe acute respiratory infection (SARI) remain significant public health burdens, particularly among older adults and in resource-limited settings. However, long-term epidemiologic trends across different pandemic phases remain poorly understood, especially for vulnerable populations.

Methods Surveillance and hospital records collected between 2017 and 2024 from two sentinel hospitals in Shanghai, China, were analyzed. Interrupted time-series (ITS) models were used to evaluate temporal changes in ILI and SARI incidence. Large-scale data processing techniques were applied to extract structured information from unstructured medical records, enabling comprehensive analysis of clinical characteristics and outcomes.

Results Both ILI and SARI cases declined sharply during the COVID-19 pandemic but rebounded thereafter. Older adults (≥60 years) experienced disproportionately greater increases in these cases, along with higher risks of ILI-to-SARI progression and ICU admission. Although clinical recovery rates improved in the post-pandemic period, the demand for oxygen therapy and hospital-based care remained elevated.

Conclusion These findings highlight the long-term impact of the COVID-19 pandemic on respiratory infection dynamics and underscore the need for targeted interventions for older adults. Despite highest susceptibility and baseline risk for severe illness, older adults demonstrated improved post-pandemic clinical outcomes due to enhanced clinical prioritization. Strengthening vaccination programs, improving health education, and enhancing resource preparedness are crucial to reduce the post-pandemic disease burden, particularly in settings with limited healthcare capacity. These findings provide important evidence to strengthen post-pandemic respiratory surveillance and improve protection strategies for older adults in China.

Keywords Influenza-like Illness, Severe Acute Respiratory Infection, COVID-19

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Introduction

Respiratory infectious diseases remain a major contributor to global morbidity and mortality, disproportionately affecting vulnerable populations such as children and older adults, particularly in regions with limited healthcare capacity. Among these diseases, influenza-like illness (ILI) and severe acute respiratory infection (SARI) are leading syndromes that account for substantial health and economic burdens worldwide [1]. The COVID-19 pandemic not only caused an unprecedented public health crisis [2–4] but also led to marked disruptions in the epidemiology of traditional respiratory pathogens [5–8]. These disruptions have raised concerns about long-term shifts in seasonal patterns, disease severity, and health system capacity.

Early evidence from diverse settings showed sharp declines in influenza and other respiratory viruses, including respiratory syncytial virus, adenovirus, and rhinovirus, during the first months of the pandemic. This was primarily attributable to nonpharmaceutical interventions such as social distancing and mobility restrictions [7–10]. For instance, a study conducted in Hong Kong reported that pediatric hospitalization rates for these respiratory viruses decreased by 85%–99% in 2020 compared with pre-pandemic levels [11]. However, as control measures eased, multiple countries reported a rebound of respiratory infections, characterized by an asynchronous pattern of resurgence. Rhinovirus rebounded first, followed by seasonal coronaviruses, RSV, and influenza viruses [12, 13]. In resource-constrained settings, where healthcare capacity is limited, the burden on older adults becomes even more pronounced, leading to more severe disease outcomes [14, 15]. Notably, most existing studies have focused on short-term trends or single pathogens, lacking long-term age-stratified analyses that integrate multi-center data. This highlights the need for further investigation into the dynamics of post-pandemic respiratory disease.

To address this gap, our study leverages eight years (2017–2024) of sentinel surveillance data from China. Interrupted time-series (ITS) models were previously used to delineate ILI and SARI trends during and after the pandemic. A special focus is placed on older adults, a demographic at elevated risk from immunosenescence, comorbidities, and barriers to healthcare. The analysis of their specific infection patterns and outcomes is intended to directly inform the development of targeted public health interventions for high-risk groups and resource-conscious settings.

Study design and methods

Data distributions and collection

Case definitions for this study adhered to the standardized protocol of Shanghai's influenza surveillance program [16]. Specifically, ILI was defined as the presence of fever (temperature ≥ 38 °C) with either cough or sore throat. In contrast, SARI was defined as an acute respiratory infection requiring hospitalization, characterized by fever (measured or self-reported temperature ≥ 38 °C) and cough, with onset within 10 days.

The study sites were located in Baoshan District, an urban–rural fringe area of Shanghai that combines densely populated urban neighborhoods with less-resourced suburban communities, reflecting heterogeneous healthcare access and service capacity. Data were obtained from two sentinel hospitals in Baoshan District, Shanghai: Wu Song Central Hospital (hereafter "Wu Center") and Baoshan Central Hospital (hereafter "Bao Center"). The study included ILI and SARI cases from January 2017 to December 2024, with extracted variables covering demographics (sex and age), key dates (visit, admission, discharge), clinical outcomes, and diagnostic results. The study period was divided into three phases: pre-pandemic (January 2017–January 2020), during-pandemic (February 2020–December 6, 2022), and post-pandemic (December 7, 2022–December 2024). Because ILI data at Bao Center were missing prior to March 2020, t-tests were conducted and confirmed no significant differences between the two hospitals' ILI data from March 2020 to December 2024. Thus, the interrupted time series analysis for ILI was performed using data solely from Wu Center.

Structured diagnostic records of ILI and SARI patients were retrieved from the hospitals' HIS and EMR systems (Fig. 1). Data collection followed a dual-path approach. First, SQL queries were used to identify potential ILI and SARI cases based on symptom-related keywords, such as "fever," yielding 113,851 suspected ILI and 11,559 suspected SARI cases. Second, a locally deployed Qwen-2-7B model, fine-tuned through knowledge distillation via a GPT-4o-based teacher-student framework and LoRA-based instruction tuning, extracted semantic information from unstructured medical histories. To ensure high extraction reliability, we implemented a multi-stage pipeline, including keyword filtering followed by dual-pass LLM screening (coarse and fine). Notably, a majority-voting strategy was employed during the fine-screening phase, where each record underwent three independent inferences to minimize classification errors. To ensure patient confidentiality, all Electronic Medical Records (EMRs) were de-identified locally, and no patient-level data were transmitted to external APIs. Validation against a clinician-audited gold standard showed a concordance rate approaching 100% (Accuracy: 98.5%, F1-score: 0.96), accurately identifying 80,007 ILI and 12,181 SARI cases. Data governance adhered to the WS445-2014 electronic medical record standard, and incorporated dynamic quality checks (integrity > 96.1%,

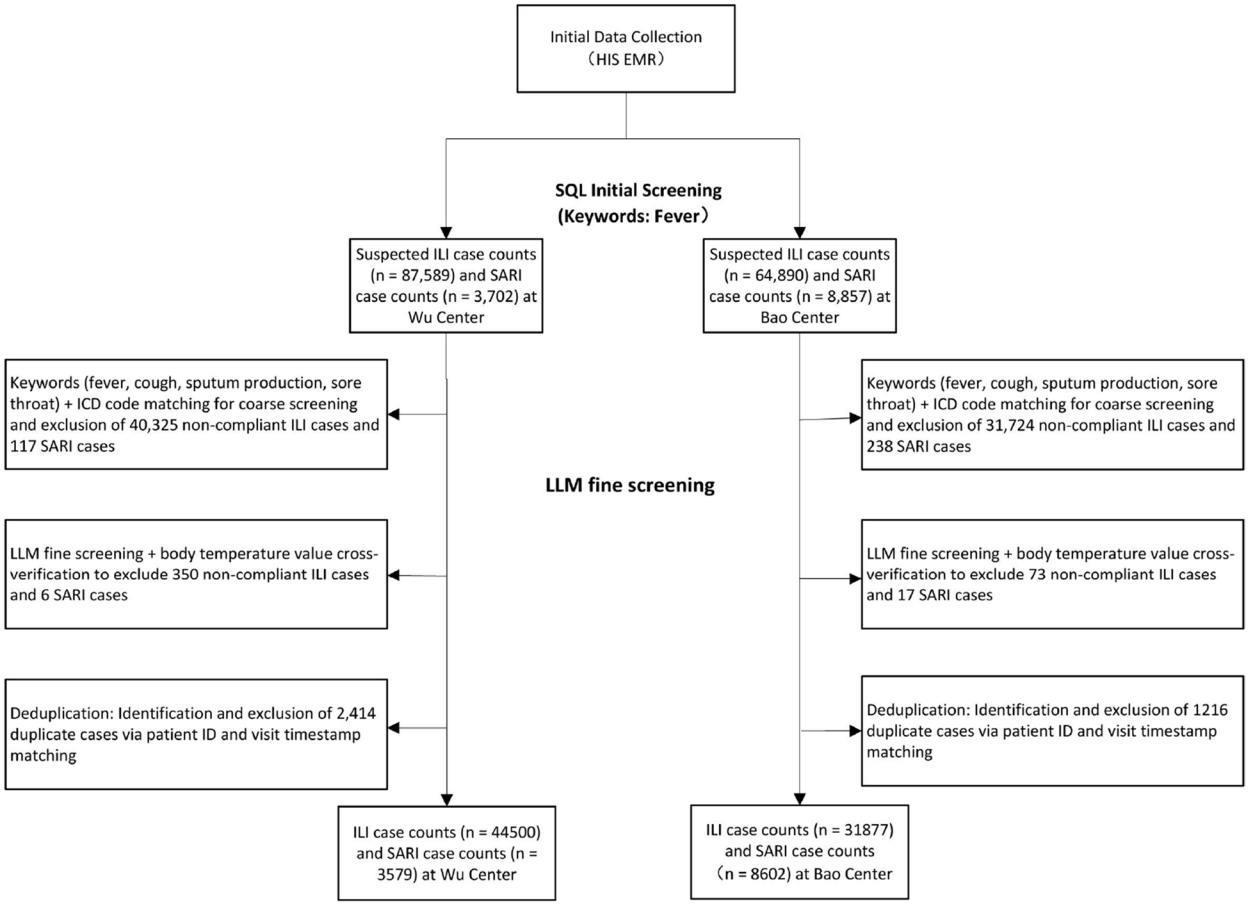


Fig. 1 Study flow chart. Flow chart showing the inclusion of patients with influenza-like illness (ILI) and severe acute respiratory infection (SARI) from 2017 to 2024. Data were derived from two sentinel hospitals in Baoshan District, Shanghai

consistency < 2.5%) and blockchain traceability, ensuring data reliability and compliance.

Ethics approval

This retrospective real-world data analysis was approved by the Clinical Trials Ethics Committee of Wu Song Central Hospital, Baoshan District, Shanghai (Approval No. Ethical Review 2024-Y-18).

ITS model

An interrupted time series regression model was employed to compare trends across the three study phases (pre-pandemic, pandemic, and post-pandemic) and to evaluate the impact of COVID-19 on trends in ILI and SARI. The model used the natural logarithm of weekly case counts as the dependent variable. The following equation was applied to analyze ILI and SARI characteristics:

$$\log(\text{cases}) = \beta^{*}\text{time} + \beta_1^{*}\text{intervention1} + \beta_2^{*}\text{postime1} + \beta_3^{*}\text{intervention2} + \beta_4^{*}\text{postime2} + \cos(2\pi/52 * time) + \sin(2\pi/52 * time)$$

Here, the model incorporated time trends, two intervention points (February 1, 2020, and December 6, 2022), and their respective post-intervention periods, along with Fourier terms (cos(2π/52∗time) and sin(2π/52∗time)) to account for seasonal variations. Statistical significance was assessed using 95% confidence intervals. First-order autocorrelation in residuals was assessed with the Durbin-Watson test (p < 0.05). When necessary, the Cochrane-Orcutt iterative method was used to correct the model [17].

Data analysis

Data organization, including checks for completeness and accuracy, was performed using Excel 2021. Descriptive statistical analyses of patient outcomes and disease severity were conducted using IBM SPSS Statistics 27. Interrupted time-series analyses were performed in R 4.4.1 (R Foundation for Statistical Computing, Vienna, Austria), where regression models were fitted to evaluate the impact of the COVID-19 pandemic on the trends of ILI and SARI.

Results

Patient characteristics

A total of 46,914 ILI patients were admitted to Wu Center, and 12,181 SARI patients were treated across Wu Center and Bao Center between 2017 and 2024 (Table 1). For ILI patients, ages ranged from newborns to 99 years, with a mean age of 34 years. The 15–59 years age group was the most prevalent, accounting for 53.4% (23,785 cases) of all ILI patients. In contrast, SARI patients had a wider age range (4 months to 100 years) and a higher mean age of 40 years, older adults (≥ 60 years) constituted the largest subgroup, accounting for 54.3% (4,889 cases) of SARI patients. Temporally, both ILI and SARI cases exhibited a consistent trend: a significant decline during the COVID-19 pandemic, followed by a marked rebound in the post-pandemic period.

ILI trends

Interrupted time-series analysis revealed a sharp decline in ILI cases during the pandemic, followed by a strong rebound in the post-pandemic period (Fig. 2). Among children (≤ 14 years), case numbers dropped most markedly during the pandemic and then partially rebounded afterward. Adults aged 15–59 years also experienced a significant decline, but their post-pandemic rebound was the most pronounced, surpassing pre-pandemic levels. In contrast, older adults (≥ 60 years) showed little change during the pandemic; however, their case numbers rose substantially after restrictions were lifted, underscoring their heightened susceptibility in the post-pandemic period.

SARI trends

Interrupted time-series analysis showed a sharp decrease in SARI cases during the COVID-19 pandemic, followed by a significant rebound in the post-pandemic period, with case counts surpassing both pandemic-era and pre-pandemic baseline levels (Fig. 3). Among children (≤ 14 years), SARI cases dropped significantly during the pandemic compared to pre-pandemic levels, then partially recovered post-pandemic, showing an apparent increase relative to the pandemic period. Adults aged 15–59 years also experienced a sharp decline in cases during the pandemic, followed by a significant rebound afterward; however, their post-pandemic levels did not differ substantially from the pre-pandemic baseline. In contrast, older adults exhibited a significant decrease in SARI cases during the pandemic. Their post-pandemic rebound was the most pronounced, with case levels surpassing both pandemic-era and pre-pandemic levels.

Clinical outcomes in older patients

Interrupted time-series analysis revealed that older patients had a significantly higher incidence of both ILI and SARI in the post-pandemic period than in the pre-pandemic baseline, underscoring their increased susceptibility. In terms of disease progression, the rates of ILI progressing to SARI and SARI requiring ICU admission

Figure 4 increased sharply during the pandemic but declined substantially afterward. In terms of clinical course and outcomes, hospitalization duration for older patients decreased in the post-pandemic period. Meanwhile, non-recovery/mortality rates declined significantly, and clinical recovery rates improved notably compared with pre-pandemic periods (χ^2 = 19.694, p < 0.001). However, oxygen therapy demand for patients with ILI progressing to SARI exceeded pre-pandemic baseline levels (p < 0.001). Overall, although clinical outcomes for older adults improved in the post-pandemic era, their healthcare resource needs, such as oxygen therapy, remained elevated relative to the pre-pandemic levels

Table 1 Demographic characteristics of study participants

Characteristics		Total	Pre-COVID		During COVID		Post-COVID	
		N	*N*	%	*N*	%	*N*	%
ILI		**44500**	**14539**	**32.7**	**6373**	**14.3**	**23588**	**52.0**
Sex	Male	21200	6942	32.7	3250	15.3	11008	51.9
	Female	23300	7597	32.6	3123	13.4	12580	56.0
Age group(years)	≤14	12637	4419	35.0	2311	18.3	5907	46.7
	15–59	23785	8036	33.8	2796	11.8	12953	54.5
	≥60	8078	2084	25.8	1266	15.7	4728	58.5
SARI		**12181**	**4645**	**38.1**	**1710**	**14.0**	**5826**	**47.8**
Sex	Male	6044	2339	38.7	901	14.9	2804	46.4
	Female	6137	2306	37.6	809	13.2	3022	49.2
Age group(years)	≤14	4851	2274	46.9	521	10.7	2056	42.4
	15–59	2431	948	39.0	318	13.1	1165	47.9
	≥60	4889	1423	29.0	871	17.8	2605	53.2

Distribution of *ILI* Influenza-like illness and *SARI* severe acute respiratory infection patients by age group, sex, and study phase (pre-COVID, During COVID, post-COVID)

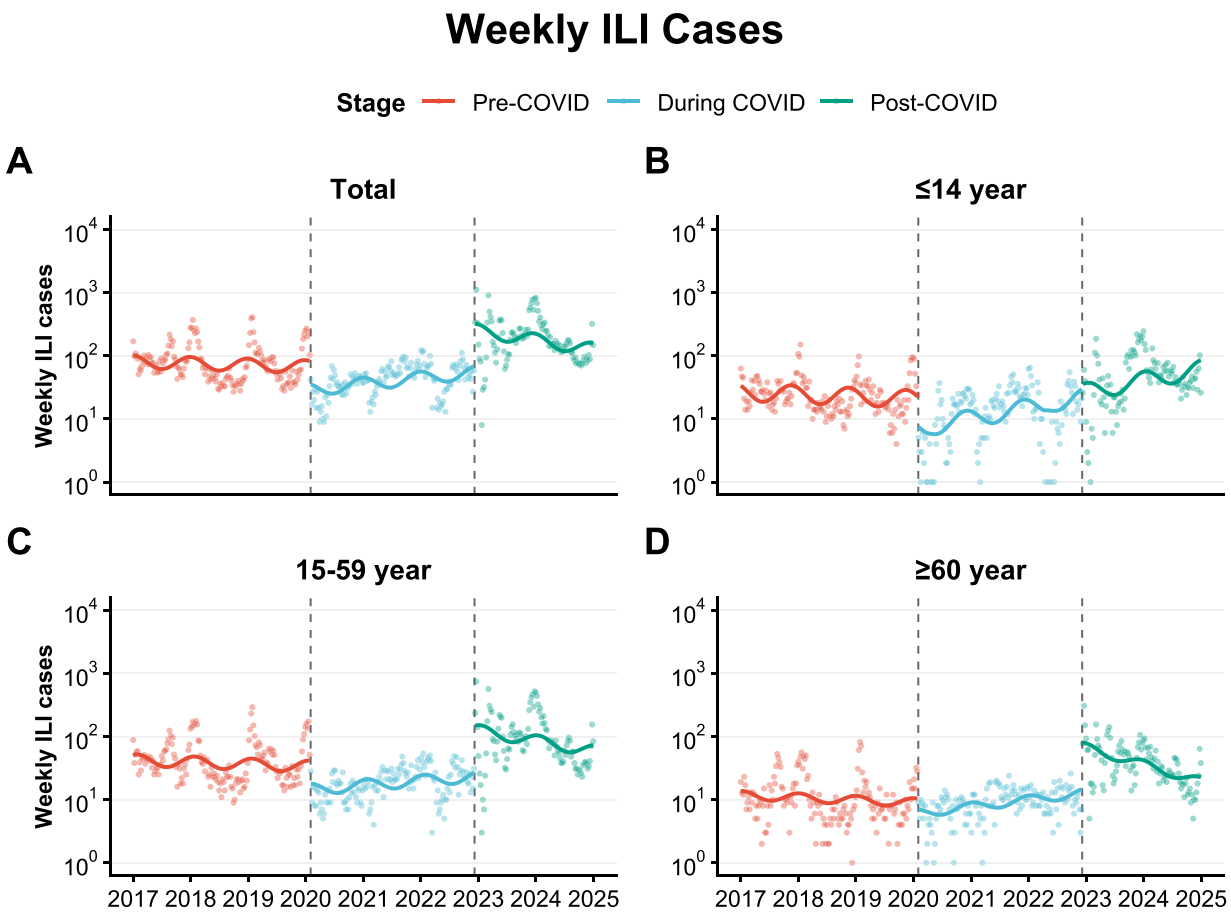


Fig. 2 Trends in influenza-like illness (ILI) by Total Population and Age Groups, 2017–2024. Weekly ILI cases stratified by age group, with each subpanel corresponding to: **A** Total population; **B** ≤ 14 years; **C** 15–59 years; **D** ≥ 60 years. Vertical dashed lines indicate the start (January 2020) and the end (December 2022) of the COVID-19 pandemic period in China. Solid lines represent observed values, while fitted regression lines illustrate interrupted time-series (ITS) estimates for pre-pandemic, pandemic, and post-pandemic phases

Discussion

This study provides a comprehensive understanding of the long-term evolution of ILI and SARI during the COVID-19 pandemic. Both ILI and SARI incidence declined substantially during the pandemic, mainly attributable to widespread preventive measures and reduced healthcare visits, followed by a clear rebound after restrictions were lifted. The extended observation period enabled the detection of delayed recovery patterns that shorter studies might have overlooked, providing a more complete picture of post-pandemic transition. In the post-pandemic phase, older adults exhibited improved clinical outcomes - including low hospitalization and ICU admission rates, as well as shortened length of hospital stays - while their consumption of medical resources such as oxygen therapy and medications remained elevated compared to pre-pandemic levels. Together, these findings suggest a structural shift in respiratory infection transmission and clinical outcomes, underscoring the importance of sustained surveillance, timely vaccination, and health system preparedness to protect aging and high-risk populations in the years following the pandemic.

Several mechanisms may account for these trends. The implementation of nonpharmaceutical interventions, such as social distancing and mask-wearing, reduced interpersonal contact and curtailed viral transmission, while decreased healthcare-seeking behavior drove lower ILI/SARI cases during the pandemic [18]. Post-pandemic social resumption and immune gaps triggered rebounds, accompanied by a resurgence in healthcare demand. Among all age groups, older adults exhibited the sharpest resurgence in infection burden, probably owing to age-related immunosenescence, a higher prevalence of chronic comorbidities, and comparatively low vaccination coverage. In contrast, their improved clinical outcomes in the post-pandemic period may reflect advances in clinical management of respiratory diseases, expanded therapeutic options, and increased prioritization of older adults in clinical care pathways [19]. These findings

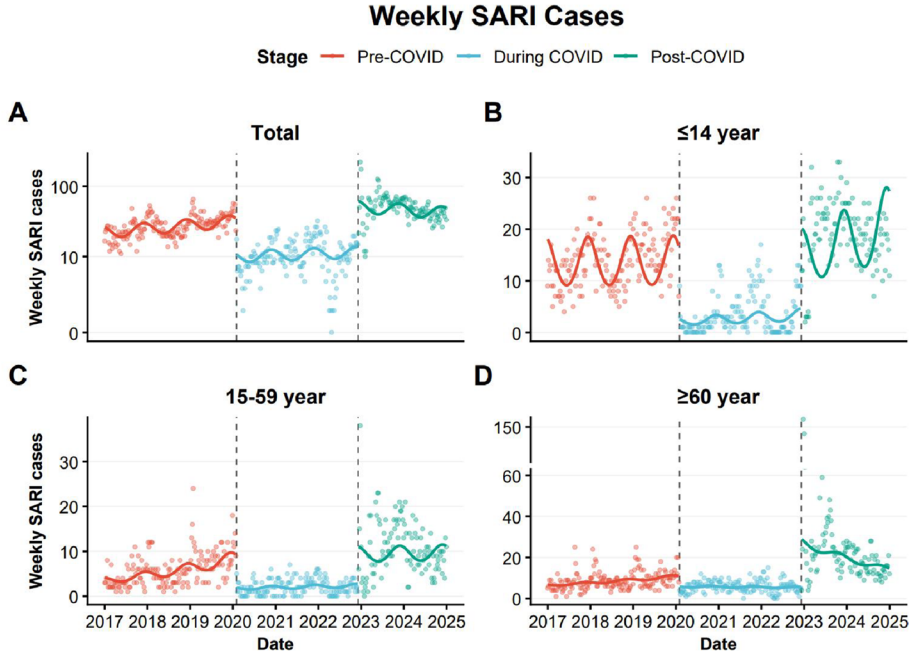
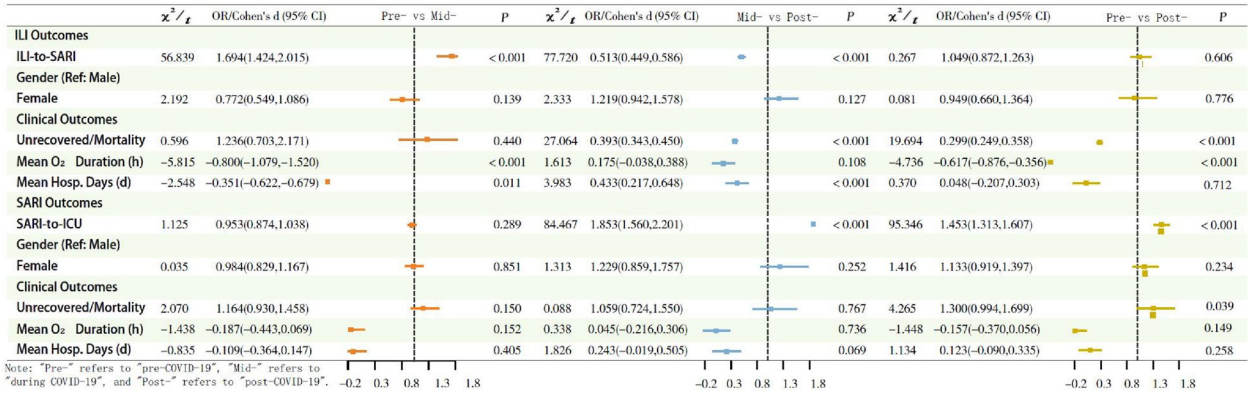


Fig. 3 Trends in severe acute respiratory infection (SARI) by Total Population and Age Groups, 2017–2024. Weekly SARI cases stratified by age group, with each subpanel corresponding to: **A** Total population; **B** ≤ 14 years; **C** 15–59 years; **D** ≥ 60 years. The y-axis is presented on a logarithmic scale to improve visualization. Vertical dashed lines mark the start and end of the pandemic period. Solid lines represent observed counts, and dotted lines show ITS fitted trends across study phases. Note: Please note that the Y-axis scales differ between panels to clearly visualize the trends despite differences in absolute case magnitudes



	χ^2/t	OR/Cohen's d (95% CI)	P	χ^2/t	OR/Cohen's d (95% CI)	P	χ^2/t	OR/Cohen's d (95% CI)	P
ILI Outcomes									
ILI-to-SARI	56.839	1.694(1.424,2.015)	< 0.001	77.720	0.513(0.449,0.586)	< 0.001	0.267	1.049(0.872,1.263)	0.606
Gender (Ref: Male)									
Female	2.192	0.772(0.549,1.086)	0.139	2.333	1.219(0.942,1.578)	0.127	0.081	0.949(0.660,1.364)	0.776
Clinical Outcomes									
Unrecovered/Mortality	0.596	1.236(0.703,2.171)	0.440	27.064	0.393(0.343,0.450)	< 0.001	19.694	0.299(0.249,0.358)	< 0.001
Mean O_2 Duration (h)	−5.815	−0.800(−1.079,−1.520)	< 0.001	1.613	0.175(−0.038,0.388)	0.108	−4.736	−0.617(−0.876,−0.356)	< 0.001
Mean Hosp. Days (d)	−2.548	−0.351(−0.622,−0.679)	0.011	3.983	0.433(0.217,0.648)	< 0.001	0.370	0.048(−0.207,0.303)	0.712
SARI Outcomes									
SARI-to-ICU	1.125	0.953(0.874,1.038)	0.289	84.467	1.853(1.560,2.201)	< 0.001	95.346	1.453(1.313,1.607)	< 0.001
Gender (Ref: Male)									
Female	0.035	0.984(0.829,1.167)	0.851	1.313	1.229(0.859,1.757)	0.252	1.416	1.133(0.919,1.397)	0.234
Clinical Outcomes									
Unrecovered/Mortality	2.070	1.164(0.930,1.458)	0.150	0.088	1.059(0.724,1.550)	0.767	4.265	1.300(0.994,1.699)	0.039
Mean O_2 Duration (h)	−1.438	−0.187(−0.443,0.069)	0.152	0.338	0.045(−0.216,0.306)	0.736	−1.448	−0.157(−0.370,0.056)	0.149
Mean Hosp. Days (d)	−0.835	−0.109(−0.364,0.147)	0.405	1.826	0.243(−0.019,0.505)	0.069	1.134	0.123(−0.090,0.335)	0.258

Fig. 4 Clinical outcomes among older adults (≥60 years) with ILI and SARI across pre-pandemic, pandemic, and post-pandemic phases. Proportions of patients progressing from ILI to SARI, requiring ICU admission, achieving clinical recovery, or experiencing death are shown for each study phase. For SARI patients, the proportions admitted to the ICU, requiring oxygen therapy, recovering, or dying are also presented, along with the mean length of hospital stay. Bars represent proportions with 95% confidence intervals

highlight the importance of targeted health education on prevention and timely treatment during flu seasons to mitigate disease burden further.

The findings of this study carry several important implications for public health and policy. First, for the older, who continue to experience greater health severity from respiratory infections than before the pandemic, targeted health education and influenza vaccination should be prioritized. Second, hospitals need to enhance preparedness for surges in demand for oxygen therapy, ICU beds, and pharmaceutical treatments through strategic stockpiling and dynamic resource scheduling. Finally, in resource-limited settings, these insights can help design cost-effective interventions. At the same time, broader efforts are needed to address the heightened health inequities exacerbated by the pandemic, ensuring that protection extends to the most vulnerable [20, 21]. The observed rebound among older adults highlights the need to prioritize vaccination, early healthcare-seeking, and hospital preparedness for this population in the post-pandemic period.

This study has notable strengths. It draws on 8 years of continuous monitoring data from two hospitals in Baoshan District, Shanghai. It adopts an innovative ITS+LLM framework for both quantitative analysis and unstructured data processing, with multi-center validation of real-world data. Meanwhile, it conducts multi-dimensional assessments of age-specific trends, clinical outcomes, and other relevant factors. The observed trends in respiratory infection cases also offer practical implications for hospital management. We recommend that healthcare institutions develop intelligent early-warning systems based on symptom-keyword monitoring to enable risk stratification and timely intervention.

This study is subject to several limitations. First, the data were obtained from two sentinel hospitals in Shanghai, and thus the findings may not be fully generalizable to differing healthcare infrastructures, notwithstanding their transferable analytical framework. Second, the lack of etiological confirmation means that ILI/SARI diagnoses relied on clinical and syndromic criteria, potentially leading to misclassification. Third, the unavailability of total testing volume precludes the calculation of positivity rates, and temporal shifts in healthcare-seeking behavior (potentially varying across age groups) during and after the pandemic might have affected case counts, although the use of standardized surveillance protocols over time helps mitigate this concern. Finally, while the interrupted time-series approach assumes relatively stable trends within each phase and may not fully account for complex nonlinear dynamics, the extended observation period and seasonal adjustments employed in the analysis enhance the validity of the findings.

Conclusion

In summary, our findings demonstrate that both ILI and SARI declined during the COVID-19 pandemic but rebounded afterward, with older adults showing the highest susceptibility and baseline risk. These results underscore the need for targeted preventive strategies, particularly vaccination and health education for the older population, as well as optimizing hospital preparedness and resource allocation. Continued surveillance and targeted interventions for older adults will be essential to reduce the post-pandemic burden of respiratory infections in China. Future research should aim to monitor the long-term sequelae of COVID-19 and assess how seasonal variations and evolving pathogens influence respiratory disease patterns.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12889-026-26616-w.

Supplementary Material 1.

Acknowledgements

We thank the medical staff at Wu Song Central Hospital, Baoshan District, Shanghai, and other participating hospitals for their support in data collection. We also acknowledge the Data Team of Fudan University for providing technical support in the large language model (LLM) application during unstructured data processing.

Authors' contributions

Y.G. (Ying Guo, First Author): Led the overall study design and coordination, drafted the initial manuscript, and revised the draft based on team feedback. N.M. (Ning Ma): Performed statistical analyses (e.g., ANOVA) and validated the accuracy of analytical results. H.X. (Huimin Xu): Collected electronic medical record data and coordinated information exchange among different teams. F.Z. (Fangyi Zhu): Assisted in the production and validation of forest plots. W.Q. (Wanyu Qiao): Guided interrupted time-series analysis methods and verified the analytical process. W.W. (Weibing Wang): Assisted in supervising the study implementation, reviewed the manuscript, and supported the finalization of the manuscript. All authors read and approved the final version of the manuscript. Y.Y. (Ye Yao): Supervised the entire study, revised the manuscript for critical intellectual content, and finalized the research direction.

Funding

This work was supported by "Construction and Application of an Early-warning Model for Acute Respiratory Infectious Diseases Based on Multi-source Data from Medical Alliances" from Wusong Central Hospital, Baoshan District, Shanghai. The funder had no role in the conception and design of the study, collection, analysis, or interpretation of data, drafting of the manuscript, or the decision to submit the manuscript for publication.

Data availability

The datasets generated and/or analyzed during the current study are not publicly available due to restrictions on the disclosure of electronic medical records (to protect patient privacy as required by Chinese healthcare regulations). However, the minimal dataset necessary to replicate the study findings is available from the first corresponding author (Ye Yao) upon reasonable request, with approval from the participating hospitals' data management committees.

Declarations

Ethics approval and consent to participate

This retrospective real-world data analysis was approved by the Clinical Trials Ethics Committee of Wu Song Central Hospital, Baoshan District, Shanghai (Approval No. Ethical Review [2024] No. 18). Given the retrospective nature of the study and the use of de-identified electronic medical record (EMR) data (no personal identifiers such as names, ID numbers, or contact information), the above-mentioned ethics committee waived the requirement for individual informed consent, in accordance with the Declaration of Helsinki.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Received: 17 December 2025 / Accepted: 6 February 2026
Published online: 24 February 2026

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